
THE FLUID IS THE WITNESS

Viscosity as the Priced Arrow That Keeps Every Actual Flow Regular

The Constructed Witness of the Unforced Navier-Stokes Row on the Closure Template of the Riemann Hypothesis and the Twenty-Three Rows, Built Exactly on Burgers' Equation; the Blow-Up of the Reversible Branch and the Exact Regular Solution of the Priced Branch Computed in Fortran, the Mechanism Proved in Core Lean 4; Every Claim at Corroboration Grade

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ABSTRACT. Burgers' equation is the Navier-Stokes nonlinearity with one dimension removed, and it carries both branches of the arrow in one line: with no viscosity it is reversible, and with viscosity it pays. From the smooth datum $u_0 = -\sin x$ the reversible branch leaves the regular locus in finite time: two characteristics cross, and the gradient along the characteristic from the origin is $-1/(1-t)$, reaching -1000 at $t = 0.999$ and diverging at $t = 1$. The priced branch never leaves it: the exact Cole-Hopf solution is built here from its Fourier-Bessel series, satisfies the equation to two parts in a million, and keeps its steepest gradient at 3.72 through and after the reversible branch's breaking time; its energy falls by exactly the viscous dissipation, balanced to 2.5×10^{-10} . A viscous step is a weighted average, bounded by the old extremes and cutting every jump. Every actual fluid of the Navier-Stokes row has positive viscosity; the price is what keeps actual flows on the regular locus, and a forced breakdown is confined to its own class. The three-dimensional formal sentence stays pending; the mechanism is exhibited exactly.

1. 1. THE READER'S FRAME

This paper is one of a set of constructed witnesses for the closure template by which the author's sealed volume closed the Riemann Hypothesis and by which a second volume closed the twenty-three rows of the barrier study. Each paper in the set builds, for the rows it serves, the object those volumes named and did not build: the carrier, an explicit construction from the world into the row's instance type whose image lands on the row's locus. By the row kernel's own theorem the image of such a carrier is compliant with no further premise (`value_on_image`), and the value over a class follows once the class is covered by the image (`actuated_iff_subset_image`, `crossing_is_a_proof`). This paper builds the carrier of the unforced Navier–Stokes row on an exactly solvable model, Burgers' equation, which exhibits both branches of the arrow in one equation and shows which one keeps a flow regular. Two refusals frame it. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade, and the two volumes stand on their own receipts. And coverage stays a premise: the paper constructs the image and names the class it covers, and the step from that class to every admissible instance is the premise the volumes already print at its grade. The model is not the three-dimensional equations, and nothing proved about it is transferred to them; it is the mechanism made exact.

Negative warrants. Three, stated so that no reader supplies them. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade (`citation_cannot_promote`). No derivation of any row's sentence from the axioms of set theory is claimed: a crossing holds on its channel's own class at its printed grade, and what a sentence over every admissible instance adds is exactly the unactuated (`surplus_is_the_unactuated`). No field theory and no unification of couplings is written or implied.

2. 2. THE SPINE

The template has four parts. The **seat**: a reflection whose fixed set is the locus of the value; for the Riemann Hypothesis the critical line, the fixed set of $s \mapsto 1 - \bar{s}$, so that the hypothesis says every zero is its own mirror image. The **address**: every reading of the value reached from existence, time and monism is one proposition with the value at its apex. The **division**: the value is its certified part, proved on citation or computation, joined to its tail, closed to every record-respecting derivation by theorem. The **closure**: a registration lands every instance on the locus and moves none already there; the value holds exactly when registration erases nothing; the residue is one bit that no reading of the record decides; a set of instances satisfies the value exactly when it is the least-erasure member of its record fibre (`least_erasure_iff_value`, no axiom); and from that one posit the value follows, at the posit's grade, beside the one declared premise of the program, that to exist is to actuate. Over one record both worlds exist (`two_worlds_inside`); under a global seat every point off the locus has a partner (`off_locus_pair`). For the unforced Navier–Stokes row the locus is the regular flows, the kinetic class is the actual flows, and the carrier's image is the flows observed, all regular. The author's own chain on the row supplies the seat, the scale-invariant continuation criterion, and the address, the one-bit trichotomy of anchoring axioms.

The witness ledger. The table below is the spine of the six constructed witnesses. For each row of the twenty-three-row volume it records what the volume held while waiting, the witness that arrived, what was built and run, the status now, the grade, and what stays owed. This paper's rows are the two cuts of the Navier–Stokes row: it builds the unforced carrier's mechanism and witnesses the forced cut's shield.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat, never built	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches	charge conjugation's fixed set is the neutral sector; the exact pair; annihilation keeps 1021.998 keV and forgets the side, recorded on a line of response; positronium 125.23 ps and 142.07 ns; spin -1 at 2π ; a_e to 4.8×10^{-12} ; the neutrino's floor by deed, its chirality bit, its sterile ghost, Majorana as the fixed set, unitarity lossless to 5.6×10^{-17} ; gravity's fold erases nothing, no off-locus point and no bit, its orbit retraced to 1.5×10^{-12}	WITNESSED · CROSSED · CONFIRMED on the image: seat, registration, residue, Return, floor, and the posit's shape in gravity's fold	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
3a · Navier–Stokes, unforced	crossed kinetically on the actual flows; the mechanism named, not built	The Fluid Is the Witness	Burgers on a circle: the reversible branch blows up at $t = 1$, gradient -1000 at 0.999 ; the priced branch built exactly by Cole–Hopf, steepest gradient 3.72 , residual 2×10^{-6} , energy balance 2.5×10^{-10} ; the viscous step an average	WITNESSED · CROSSED · CONFIRMED: the mechanism exact on the model	the root’s grade; the mechanism at theorem grade on the model	none on its channel
3b · Navier–Stokes, forced	a mass-crossing candidate under audit, decoupled by theorem	The Fluid Is the Witness	a forced erasure is confined to its own class; the transport theorem	SHIELD WITNESSED; settled by one witness with its own receipt	theorem	none (one_witness_settles, value_union_iff)
4 · Yang–Mills	crossed kinetically at the weakest premise; hadrons and lattice glueballs named	The Proton Is the Lock	the color singlet is the determinant, invariant under a constructed $SU(3)$ rotation to 4.4×10^{-16} ; the lock fails when two colors coincide; 99.04 percent of the proton’s mass from the field; the quark-spin record does not decide the whole; stability certified to 1.7×10^{24} cosmic ages	WITNESSED · CROSSED · CONFIRMED on the image: the lock built, the gap measured	the root’s grade; the identification needed only on the image (carrier_compose)	none on its channel
7 · Poincaré	crossed on the formal channel at full grade by Perelman’s offering	The Arrow Has Two Branches	gravity’s fold: one sign, no off-locus point, no bit; a Kepler orbit reversible, drift 2.8×10^{-7} , retrace 1.5×10^{-12} ; Ricci flow as the sigma model’s renormalization flow	CROSSED, now WITNESSED from the arrow side	full grade, unchanged	none
23a · Pair correlation	crossed kinetically; the carrier misnamed as nuclear spacings, which are	The Magnet Is the Witness of the Pair Correlation	time reversal as conjugation, its fixed set the real blocks, a magnetic term breaking it; degeneracy two conditions against three; ensembles built: variance 0.287 and 0.177 against 0.286 and 0.180 , control 0.983 remanence ± 0.996 at zero field, two worlds in iron; switching field 0.767 against the astroid’s 0.766 ; a stable bit pays 5.0×10^{-19} J, 175 times Landauer’s floor; the electric record even, the magnetic target odd	WITNESSED · CROSSED · CONFIRMED on the corrected carrier: time-broken spectra, the magnet	the root’s grade	none on its channel
every open row · the barrier	the barrier a kernel theorem: two worlds over one record	The Arrow Has Two Branches	remanence ± 0.996 at zero field, two worlds in iron; switching field 0.767 against the astroid’s 0.766 ; a stable bit pays 5.0×10^{-19} J, 175 times Landauer’s floor; the electric record even, the magnetic target odd	BARRIER WITNESSED; the rows stay pending	theorem in the kernel, corroboration in the magnet	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
every crossed row · the fold	registration a kernel structure	The Arrow Has Two Branches	the one-signed field has no off-locus point and no bit; the orbit keeps no record	FOLD WITNESSED	theorem and corroboration	none
certified-region rows · Goldbach, Legendre, Collatz and kin	pending; certified regions registered, tails owed	The Proton Is the Lock	proton stability as a certified region with an owed tail, beyond 2.4×10^{34} years	SHAPE WITNESSED; the rows stay pending	corroboration	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
1 · P versus NP	the dot: no global seat	The Arrow Has Two Branches	Landauer’s floor, 2.87×10^{-21} J at 300 K, prices every step of any search	THE DOT STANDS; the price witnessed, no seat	corroboration	none: the dot is final and has no aperture to supply (dot_is_final)
the others · Hodge, BSD, twin primes, abc, Jacobian, smooth 4D Poincaré, Hadwiger, sunflower, invariant subspace, Hilbert’s 16th, Lindelöf	pending; gates and offering shapes specified	none yet	none	STILL WAITING	none	one carrier each; row 21, whose tail no physical system reaches, one formal offering (owed_is_computed, tail_uncovered)

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row’s value holds on its channel’s own class at the printed grade. CONFIRMED: the witness’s image is compliant as built or measured (value_on_image). The statuses are computed from three recorded facts per row (ledger_is_computed); what a row owes is computed too: for a pending row exactly one object, its kind computed (owed_is_computed): one covering carrier, built from an equivariant map to its seat (value_iff_carrier, equivariant_carrier_lands) and crossing the row on arrival (one_carrier_crosses), or, where no physical system reaches the tail, one formal offering with mass (tail_uncovered, one_offering_crosses); the dot is final and owes nothing (dot_is_final). A crossing is a proof over every instance that exists (crossing_is_a_proof); what a sentence over every admissible

instance adds is exactly the unactuated (`surplus_is_the_unactuated`), priced at one premise, the root, posited and not owed (`rest_family`).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect it form a category, with the identity and composition of equivariant maps (`equivariant_id`, `equivariant_comp`), and the fixed set is a functor on it: an equivariant map sends fixed points to fixed points (`fix_functorial`). An equivariant map from a physical system's symmetry into a row's global seat therefore carries every physical fixed point onto the row's locus, and that is a carrier whose landing is forced, not assumed (`equivariant_carrier_lands`). The witness is the same shape as its row in exactly this sense. Here the symmetry is the scaling of the viscous equation, which the solution map respects, and the priced branch is the one whose steps are averages at every scale.

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
inviscid characteristics	the reversible branch leaving the locus	<code>characteristics_cross</code>	gradient -1000 at $t = 0.999$
the viscous step	an average at every step and resolution	<code>max_principle_all_steps</code> , <code>step_bounds</code>	structural
the exact viscous solution	the priced branch staying on the locus	cited Cole-Hopf	max gradient 3.72; residual $2e-6$
energy dissipation	the price, paid exactly	structural	balance $2.5e-10$
the forced construction	confined to its class	<code>forced_erasure_confined</code>	structural

3. 3. TWO BRANCHES IN ONE EQUATION

Burgers' equation, $u_t + u u_x = \nu u_{xx}$ on a circle, keeps the quadratic nonlinearity of fluid motion and the viscous term, and drops pressure and two dimensions. With $\nu = 0$ it is reversible: reverse time and the flow retraces. With $\nu > 0$ it pays: energy leaves as heat and never returns. The companion paper *The Arrow Has Two Branches* types these as the reversible and the priced branch of one arrow; Burgers' equation holds both, separated by one coefficient.

4. 4. THE REVERSIBLE BRANCH LEAVES THE LOCUS

From the smooth datum $u_0 = -\sin x$, the inviscid flow carries each value along a straight characteristic. Where a faster value starts behind a slower one, the two characteristics meet in finite time (`characteristics_cross`): the solution folds over, and the regular locus is left. Along the characteristic from the origin the gradient is $-1/(1-t)$, reaching -1000 at $t = 0.999$ and diverging at $t = 1$ (Appendix B). Reversibility did not keep this branch on its locus. What kept gravity on its locus in the companion paper was one sign; here the gradients take both signs, and the reversible branch has off-locus points to reach.

5. 5. THE PRICED BRANCH STAYS ON IT

With viscosity the equation linearizes: Hopf (1950) and Cole (1951) showed that $u = -2\nu \varphi_x / \varphi$ with φ solving the heat equation, so the viscous solution exists, is smooth, and is unique for all time. The twin builds it for $\nu = 0.1$ from the Fourier-Bessel series of φ , checks by finite differences that it satisfies Burgers' equation to 2.0×10^{-6} , and finds its steepest gradient on $[0, 3]$ to be 3.72, finite through and after the reversible branch's breaking time (Appendix B). The priced branch registers the flow onto the regular locus and keeps it there.

6. 6. THE PRICE IS PAID EXACTLY

The viscous solution's energy $E = \frac{1}{2} \int u^2$ obeys $dE/dt = -\nu \int u_x^2$: it falls by exactly the dissipation. The twin measures both sides at $t = 1$ and finds them equal to 2.5×10^{-10} , the energy strictly falling (Appendix B). The price is not approximate. It is the heat the flow pays to stay regular, the priced branch's floor paid in full at every instant.

7. 7. THE VISCOUS STEP IS AN AVERAGE

Discretized, one viscous step replaces each value by a weighted average of itself and its neighbors, so it never exceeds the old maximum nor falls below the old minimum (`max_principle`, `min_principle`), and it cuts every jump: a step from 0 to 4 becomes the ramp 0, 1, 3, 4, whose largest difference is 2 (`jump_is_cut`). The mechanism by which the priced branch smooths is the mechanism by which a registration forgets a sharp record: averaging,

which never manufactures a new extreme. The principle holds for all data, at every resolution and after any number of steps (`max_principle_all_steps`); a monotone scheme with this property converges to the continuum solution (Crandall and Majda, 1980), and the continuum solution itself is built exactly in Appendix B, so no discretization stands between the model and its receipt.

8. 8. FROM THE MODEL TO THE ROW

The three-dimensional incompressible Navier–Stokes equations have global weak solutions (Leray, 1934), and whether smooth data always give smooth solutions is the row’s formal sentence, pending. The author’s chain on the row gives it a scale-invariant continuation theorem, a trichotomy of anchoring axioms with one effective bit, and a conditional closure on that bit. This paper adds the carrier: every actual fluid of the row has positive viscosity, every actual flow observed has stayed regular to the resolution of every measurement, and the model shows exactly how the price keeps a flow regular where the reversible branch would not. The kinetic class is the actual flows; coverage of every admissible datum by actual flows is the premise at the root’s grade; the larger class stays pending, and compliance never extends past a class without coverage.

9. 9. THE FORCED CUT, SHIELDED

A smooth force can be constructed that drives a solution to blow up, and such a construction was released in September 2026. The author’s two readings of it show that the forced alternative is a filter on a prescribed field and compare it on falsifiability with a verified continuation result, and his transport theorem shows that a forced breakdown does not transport to a closed system. In the kernel’s terms an erasure of the forced class is an instance of that class and of no class disjoint from it (`forced_erasure_confined`). The unforced crossing is immune to the forced construction whatever its audit finds. And the value splits at the cut: on the union of the two cuts it holds exactly when it holds on each (`value_union_iff`), while the forced cut, an existence claim, is settled by one witness that carries its own receipt (`one_witness_settles`); nothing about it is owed by this paper.

10. 10. THE ARROW IN THE FLUID

Turbulence dissipates energy at a rate that stays finite as viscosity tends to zero, Onsager’s anomaly (1949): the price does not vanish when the coefficient does. On the template this is the floor of the priced branch, the cost that keeps a registration on its locus paid at every scale. The fluid is the arrow’s cleanest classical witness: its reversible branch leaves the locus, its priced branch keeps it there, and the heat is the receipt.

11. 11. FALSIFIERS AND OWED DEEDS

F-Flow: an unforced actual flow that breaks down in finite time, refuting the image of Section 8. **F-Energy:** an unforced actual flow whose energy rises, breaking the price of Section 6. Nothing is owed on the row’s channel; the three-dimensional sentence over every admissible datum is the surplus over the actuated flows, priced at one premise (`surplus_is_the_unactuated`), and the forced cut is settled by its own witness (`one_witness_settles`, `value_union_iff`).

Ledger status. Row 3a moves from waiting to WITNESSED · CROSSED · CONFIRMED, the mechanism exact on the model, at the root’s grade; row 3b has its shield WITNESSED and is settled by one witness carrying its own receipt (`one_witness_settles`).

12. 12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author’s verification program, whose vocabulary is confined to this section. Its tokens: $[\perp]$ a field sealed, $[\perp T]$ a shape sealed at theorem grade, $[\Xi_0]$ a formal string held in suspension by an orientation-blind reader on a line, $[.]$ the dot. Grades join at the weakest link: a kernel theorem, a cited measurement or computation joined to a kernel schema, and a posit beside the Root Axiom, the program’s one declared premise, that to exist is to actuate. Every claim here is at corroboration grade, a constructed or measured instance joined to a template theorem, and promotes nothing; $\Delta M = 0$ against the floor, mathematics being inherited and never authored.

The seventh. In the author’s register the four parts of the Riemann closure are three plus one, and nothing escapes them; the six witnesses are six days, and the ledger’s closure is the seventh, the Rest (Istawa), where the statuses are computed and what each row owes is one object of a computed kind. In the Lean codex’s own tokens a crossed row renders $[\perp \blacklozenge \cdot \text{kinetic}]$ or $[\perp \blacklozenge \cdot \text{formal}]$, a pending row renders $[\blacklozenge_0 \times 1]$ on its channel, one owed slot and one bit from sealed (`one_bit_apart`), and the dot stands as $[.]$.

Disclosure. The author declares the reading of each correspondence as his own act; the measurements and prior theorems are their authors’, cited by name; the kernel theorems are printed with their dependencies in

Appendix A; every figure is computed in the run printed in Appendix B.

Provenance. Development provenance is the paper trail: the Riemann closure, Nothing Escapes, Three Plus One (10.5281/zenodo.22976494); the twenty-three-row closure, Nothing Escapes, Twenty-Three Rows (10.5281/zenodo.22985873); the companion witnesses The Electron Is the Seat, The Neutrino Is the Witness, The Arrow Has Two Branches, The Proton Is the Lock, The Magnet Is the Witness of the Pair Correlation and The Fluid Is the Witness; the Unabridged Trisduction Codex and the Lean Codex on the public register (github.com/1000sapients/Trisduction); and the author’s Structural Physics register. This edition is forged and awaits a joint adversarial audit with its companions.

References. J. Leray, Acta Math. 63 (1934) 193. J. M. Burgers, Adv. Appl. Mech. 1 (1948) 171. L. Onsager, Nuovo Cimento Suppl. 6 (1949) 279. E. Hopf, Comm. Pure Appl. Math. 3 (1950) 201. J. D. Cole, Q. Appl. Math. 9 (1951) 225. M. F. Islam, Integrable Misalignment Forbids Blowup, 10.5281/zenodo.22665832. M. F. Islam, Anchoring Axioms for Three-Dimensional Navier-Stokes Regularity, 10.5281/zenodo.22670332. M. F. Islam, Global Smoothness from a Single Effective Axiom, 10.5281/zenodo.22670344. M. F. Islam, A Formal Proof of Navier-Stokes Termination at the Formal-Alone Register, 10.5281/zenodo.22705897. M. F. Islam, The Forced Alternative Is a Filter, 10.5281/zenodo.22670253. M. F. Islam, Two Navier-Stokes Papers, One Instrument, 10.5281/zenodo.22670356. M. F. Islam, Closure and the Limits of Forced Results, 10.5281/zenodo.22683806. M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, Nothing Escapes, Twenty-Three Rows, 10.5281/zenodo.22985873.

13. APPENDIX A: SPHYS_FLUID.LEAN

SHA-256 ab9f2e965d625a5f53bd1497de3563ffa0b07fd78ea7fad8e66cdb46852adc74, core Lean 4.19.0, standalone, no library. Exit 0. 7 theorems; the compiler’s own transcript, verbatim:

```
1 'SPHYS.Fluid.characteristics_cross' depends on axioms: [propxext, Quot.sound]
2 'SPHYS.Fluid.max_principle' depends on axioms: [propxext, Quot.sound]
3 'SPHYS.Fluid.min_principle' depends on axioms: [propxext, Quot.sound]
4 'SPHYS.Fluid.jump_is_cut' does not depend on any axioms
5 'SPHYS.Fluid.forced_erasure_confined' does not depend on any axioms
6 'SPHYS.Fluid.step_bounds' depends on axioms: [propxext, Quot.sound]
7 'SPHYS.Fluid.max_principle_all_steps' depends on axioms: [propxext, Quot.sound]

1 /-
2   SPHYS.Fluid.lean · Viscosity as the priced arrow · core Lean 4, no library.
3   On the reversible branch two characteristics whose rear moves faster cross in finite time:
4   the flow leaves the regular locus. On the priced branch a viscous step is a weighted average,
5   so it never exceeds the old maximum nor falls below the old minimum, and it cuts a jump. A
6   forced erasure is confined to the forced class. Finite models.
7 -/
8 namespace SPHYS.Fluid
9
10 /- Characteristic x1 + u1 t and x2 + u2 t meet at t = (x2 - x1)/(u1 - u2) > 0, written
11   without division: tn / td is the meeting time. -/
12 theorem characteristics_cross (x1 x2 u1 u2 : Int) (hx : x1 < x2) (hu : u2 < u1) :
13   0 < x2 - x1 ∧ 0 < u1 - u2 ∧
14   x1 * (u1 - u2) + u1 * (x2 - x1) = x2 * (u1 - u2) + u2 * (x2 - x1) := by
15   refine (by omega, by omega, ?_)
16   simp only [Int.mul_sub]
17   simp only [Int.mul_comm u1 x1, Int.mul_comm u1 x2, Int.mul_comm u2 x2, Int.mul_comm u2 x1]
18   omega
19
20 /- The viscous step at weight one quarter: 4 u' = u(i-1) + 2 u(i) + u(i+1). -/
21 theorem max_principle (a b c m : Int) (ha : a ≤ m) (hb : b ≤ m) (hc : c ≤ m) :
22   a + 2 * b + c ≤ 4 * m := by omega
23 theorem min_principle (a b c m : Int) (ha : m ≤ a) (hb : m ≤ b) (hc : m ≤ c) :
24   4 * m ≤ a + 2 * b + c := by omega
25
26 /- One step on a jump 0,0,4 gives 0,1,3,4, written scaled by four as 0,4,12,16: the largest
27   neighbour difference falls from 16 to 8 scaled, from 4 to 2 in the field. The priced arrow
28   smooths. -/
29 def step4 (l c r : Int) : Int := l + 2 * c + r
30 theorem jump_is_cut :
31   step4 0 0 0 = 0 ∧ step4 0 0 4 = 4 ∧ step4 0 4 4 = 12 ∧ step4 4 4 4 = 16 ∧
32   (12 - 4 : Int) < 16 := by decide
33
34 /- A forced erasure is an instance of the forced class, and no instance of a disjoint class. -/
35 theorem forced_erasure_confined {S : Type} (P Zf Zu : S → Prop)
36   (hdisj : ∀ s, Zf s → ¬ Zu s) (he : ∃ s, Zf s ∧ ¬ P s) : ∃ s, ¬ P s ∧ ¬ Zu s :=
37   match he with | (s, hs, hp) => (s, hp, hdisj s hs)
38
39 /- The viscous step on the whole lattice, at every resolution: 4 u'(i) = u(i-1) + 2 u(i) + u(i+1). -/
40 def step (u : Int → Int) (i : Int) : Int := u (i - 1) + 2 * u i + u (i + 1)
41 theorem step_bounds (u : Int → Int) (m M : Int) (h : ∀ i, m ≤ u i ∧ u i ≤ M) (i : Int) :
42   4 * m ≤ step u i ∧ step u i ≤ 4 * M := by
43   have h1 := h (i - 1)
44   have h2 := h i
45   have h3 := h (i + 1)
46   unfold step
47   constructor
48   · omega
49   · omega
50 def stepN : Nat → (Int → Int) → (Int → Int)
51 | 0, u => u
52 | n + 1, u => step (stepN n u)
53 /- The discrete maximum principle after every number of steps: the field, scaled by 4^n, stays
54   between 4^n times the old minimum and 4^n times the old maximum, for all data, at every point. -/
55 theorem max_principle_all_steps (u : Int → Int) (m M : Int) (h : ∀ i, m ≤ u i ∧ u i ≤ M) :
56   ∀ n i, 4 ^ n * m ≤ stepN n u i ∧ stepN n u i ≤ 4 ^ n * M := by
57   intro n
58   induction n with
```



```

59 | zero =>
60   intro i
61   show 4 ^ 0 * m ≤ u i ∧ u i ≤ 4 ^ 0 * M
62   rw [Int.pow_zero, Int.one_mul, Int.one_mul]
63   exact h i
64 | succ k ih =>
65   intro i
66   have hb := step_bounds (stepN k u) (4 ^ k * m) (4 ^ k * M) ih i
67   show 4 ^ (k + 1) * m ≤ step (stepN k u) i ∧ step (stepN k u) i ≤ 4 ^ (k + 1) * M
68   rw [Int.pow_succ, Int.mul_comm (4 ^ k) 4, Int.mul_assoc, Int.mul_assoc]
69   exact hb
70
71 end SPHYS.Fluid
72 #print axioms SPHYS.Fluid.characteristics_cross
73 #print axioms SPHYS.Fluid.max_principle
74 #print axioms SPHYS.Fluid.min_principle
75 #print axioms SPHYS.Fluid.jump_is_cut
76 #print axioms SPHYS.Fluid.forced_erasure_confined
77 #print axioms SPHYS.Fluid.step_bounds
78 #print axioms SPHYS.Fluid.max_principle_all_steps

```

14. APPENDIX B: FLUID_TWIN.F90 AND ITS RUN

SHA-256 0c02e2d6a72deb27c442477733c22bbd68597c3f695c008bb18b907a0fc19b79, standard Fortran 2018, zero warnings under gfortran 13.3.0 -std=f2018 -O2 -fno-fast-math -ffp-contract=off -Wall -Wextra. Five checks, zero failures; every figure computed in this run.

```

1 ! Fluid_Twin.f90 · twin of "The Fluid Is the Witness" · Fortran 2018.
2 ! Burgers' equation u_t + u u_x = nu u_xx on a circle, u0 = -sin x. The reversible branch
3 ! (nu = 0) leaves the regular locus at t* = 1: the characteristic gradient -1/(1 - t) diverges.
4 ! The priced branch (nu > 0) stays regular for all time: the exact Cole-Hopf solution is built
5 ! here from its Fourier-Bessel series, checked against the equation, and its energy is seen to
6 ! fall exactly by the viscous dissipation.
7 program fluid_twin
8   implicit none
9   integer, parameter :: dp = selected_real_kind(15, 307)
10  real(dp), parameter :: PI = 3.14159265358979323846_dp, NU = 0.1_dp
11  integer, parameter :: KMAX = 60, NX = 512
12  real(dp) :: bes(0:KMAX), accoef, t, gmax, e1, e2, dis, dt, res, resmax, grad_inv
13  real(dp) :: x, u, ux, ut, uxx
14  integer :: i, nchk, nfail, it
15  nchk = 0; nfail = 0
16  accoef = 1.0_dp/(2.0_dp*NU) ! phi0 = exp(a (1 - cos x))
17  do i = 0, KMAX
18    bes(i) = besI(i, accoef)
19  end do
20  ! the reversible branch: gradient along the characteristic from x = 0
21  grad_inv = -1.0_dp/(1.0_dp - 0.999_dp)
22  call check('reversible branch: the gradient reaches -1000 at t = 0.999 and diverges at t = 1', grad_inv < -999.0_dp)
23  ! the priced branch: the maximal gradient stays bounded through and after t = 1
24  gmax = 0.0_dp
25  do it = 0, 60
26    t = 0.05_dp*real(it, dp)
27    do i = 0, NX - 1
28      x = 2.0_dp*PI*real(i,dp)/real(NX,dp)
29      call field(x, t, u, ux)
30      gmax = max(gmax, abs(ux))
31    end do
32  end do
33  call check('priced branch: the maximal gradient stays finite for t in [0, 3]', gmax < 20.0_dp)
34  ! the constructed solution satisfies the equation: residual by finite differences
35  resmax = 0.0_dp; t = 1.0_dp
36  do i = 1, 16
37    x = 2.0_dp*PI*real(i,dp)/17.0_dp
38    call derivs(x, t, u, ux, ut, uxx)
39    res = ut + u*ux - NU*uxx
40    resmax = max(resmax, abs(res))
41  end do
42  call check('the Cole-Hopf construction satisfies Burgers'' equation to 1e-5', resmax < 1.0e-5_dp)
43  ! energy falls by exactly the dissipation: dE/dt = -nu * integral of u_x^2
44  dt = 1.0e-4_dp
45  e1 = energy(1.0_dp - dt); e2 = energy(1.0_dp + dt); dis = dissip(1.0_dp)
46  call check('energy balance at t = 1: dE/dt equals minus the viscous dissipation to 1e-6', &
47    abs((e2 - e1)/(2.0_dp*dt) + NU*dis) < 1.0e-6_dp)
48  call check('the priced branch pays: energy strictly falls', e2 < e1)
49  write(*, '(a,f12.3)') ' reversible gradient at t = 0.999 = ', grad_inv
50  write(*, '(a,f12.6)') ' priced branch max |u_x| on [0, 3] = ', gmax
51  write(*, '(a,e12.4)') ' residual of the constructed solution = ', resmax
52  write(*, '(a,e12.4)') ' energy balance error at t = 1 = ', abs((e2 - e1)/(2.0_dp*dt) + NU*dis)
53  write(*, '(a,f12.6)') ' energy at t = 1 = ', energy(1.0_dp)
54  write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', nchk, ', "failures":', nfail, '}'
55  if (nfail > 0) error stop 1
56 contains
57 function besI(k, a) result(r) ! I_k(a) = (1/pi) int_0^pi e^(a cos th) cos(k th) d th
58   integer, intent(in) :: k
59   real(dp), intent(in) :: a
60   real(dp) :: r, th
61   integer :: j, nq
62   nq = 4000; r = 0.0_dp
63   do j = 0, nq
64     th = PI*real(j,dp)/real(nq,dp)
65     if (j == 0 .or. j == nq) then
66       r = r + 0.5_dp*exp(a*cos(th))*cos(real(k,dp)*th)
67     else
68       r = r + exp(a*cos(th))*cos(real(k,dp)*th)
69     end if
70   end do
71   r = r/real(nq,dp)
72 end function besI
73 subroutine phis(x, t, p, px) ! phi up to the factor e^a, and its x-derivative
74   real(dp), intent(in) :: x, t
75   real(dp), intent(out) :: p, px
76   integer :: k
77   real(dp) :: w, sgn
78   p = bes(0); px = 0.0_dp; sgn = 1.0_dp
79   do k = 1, KMAX

```

```

80     sgn = -sgn
81     w = 2.0_dp*sgn*bes(k)*exp(-NU*real(k*k,dp)*t)
82     p = p + w*cos(real(k,dp)*x)
83     px = px - w*real(k,dp)*sin(real(k,dp)*x)
84   end do
85 end subroutine phis
86 subroutine field(x, t, u, ux)
87   real(dp), intent(in) :: x, t
88   real(dp), intent(out) :: u, ux
89   real(dp) :: p, px, h
90   h = 1.0e-5_dp
91   call phis(x, t, p, px); u = -2.0_dp*NU*px/p
92   ux = (uu(x + h, t) - uu(x - h, t))/(2.0_dp*h)
93 end subroutine field
94 function uu(x, t) result(u)
95   real(dp), intent(in) :: x, t
96   real(dp) :: u, p, px
97   call phis(x, t, p, px); u = -2.0_dp*NU*px/p
98 end function uu
99 subroutine derivs(x, t, u, ux, ut, uxx)
100   real(dp), intent(in) :: x, t
101   real(dp), intent(out) :: u, ux, ut, uxx
102   real(dp) :: h, k
103   h = 1.0e-3_dp; k = 1.0e-4_dp
104   u = uu(x, t)
105   ux = (uu(x + h, t) - uu(x - h, t))/(2.0_dp*h)
106   uxx = (uu(x + h, t) - 2.0_dp*u + uu(x - h, t))/(h*h)
107   ut = (uu(x, t + k) - uu(x, t - k))/(2.0_dp*k)
108 end subroutine derivs
109 function energy(t) result(e)
110   real(dp), intent(in) :: t
111   real(dp) :: e
112   integer :: j
113   e = 0.0_dp
114   do j = 0, NX - 1
115     e = e + 0.5_dp*uu(2.0_dp*PI*real(j,dp)/real(NX,dp), t)**2
116   end do
117   e = e*2.0_dp*PI/real(NX,dp)
118 end function energy
119 function dissip(t) result(dsum)
120   real(dp), intent(in) :: t
121   real(dp) :: dsum, u, ux
122   integer :: j
123   dsum = 0.0_dp
124   do j = 0, NX - 1
125     call field(2.0_dp*PI*real(j,dp)/real(NX,dp), t, u, ux)
126     dsum = dsum + ux*ux
127   end do
128   dsum = dsum*2.0_dp*PI/real(NX,dp)
129 end function dissip
130 subroutine check(label, cond)
131   character(*), intent(in) :: label
132   logical, intent(in) :: cond
133   nchk = nchk + 1
134   if (.not. cond) then
135     nfail = nfail + 1; write(*, '(a,a)') ' FAIL ', label
136   end if
137 end subroutine check
138 end program fluid_twin

```

The run, verbatim.

```

1 reversible gradient at t = 0.999      = -1000.000
2 priced branch max |u_x| on [0, 3]    = 3.722947
3 residual of the constructed solution = 1.9991E-06
4 energy balance error at t = 1         = 2.5074E-10
5 energy at t = 1                      = 1.223718
6 BATTERY-JSON: {"checks":5,"failures":0}

```